

After class: check your understanding

Dr. NAM · Faculty of Pharmacy, UiTM Bertam, Penang

About 25 minutes. Questions follow the six parts of the lecture and end with a pharmacy scenario.

Answer without notes first, then use the slides to check. Use the atomic masses given in each question. A periodic table and the answer key are at the back.

Part 1 · Energy changes in reactions

Q1 CR2-LO1

In an exothermic reaction:

- A. the bonds formed release more energy than the bonds broken absorb
- B. the bonds broken absorb more energy than the bonds formed release
- C. no bonds are broken
- D. the surroundings get colder

Q2 CR2-LO1

$2\text{H}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{H}_2\text{O}(\ell)$, $\Delta H = -572 \text{ kJ}$. What is ΔH for $\text{H}_2(\text{g}) + \frac{1}{2}\text{O}_2(\text{g}) \rightarrow \text{H}_2\text{O}(\ell)$? Include the sign.

Q3 CR2-LO1

$\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$, $\Delta H = -890 \text{ kJ}$. How much heat is released when 8.00 g of CH_4 burns? (C = 12.01, H = 1.01)

Q4 CR2-LO1

A cold pack contains 20.0 g of NH_4NO_3 (80.06 g/mol). Dissolving it has $\Delta H = +25.7$ kJ/mol. How much heat does it absorb?

Q5 CR2-LO1

$\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 \rightarrow 6\text{CO}_2 + 6\text{H}_2\text{O}$, $\Delta H = -2803$ kJ. How much heat does the body release by oxidising 4.50 g of glucose (180.18 g/mol)?

Part 2 • Energy diagrams

An energy diagram shows reactants at 60 kJ, the transition state at 160 kJ and products at 25 kJ.

Q6 CR2-LO2

E_a for the forward reaction:

Q7 CR2-LO2

ΔH for the reaction (include the sign):

Q8 CR2-LO2

E_a for the reverse reaction:

Q9 CR2-LO2

Match each feature of an energy diagram.

Options: transition state · E_a · ΔH · exothermic

the top of the hill

peak – reactants

products – reactants

products below reactants

Part 3 · Collision theory

Q10 CR2-LO3

Will each collision form products?

Options: effective · ineffective

Enough energy, correct orientation

Enough energy, wrong orientation

Too little energy, correct orientation

Q11 CR2-LO3

A reaction with a large E_a is slow because:

- A. its particles never collide
- B. only a small fraction of collisions have energy $\geq E_a$
- C. it is always endothermic
- D. its ΔH is large

Part 4 · Reaction rates

Q12 CR2-LO4

In a reaction, [A] falls from 0.50 M to 0.30 M in 40 s. What is the average rate of loss of A? Give the number in units of 10^{-3} M/s.

Q13 CR2-LO4

For $2\text{H}_2\text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{O}_2$, H_2O_2 is used up at $6.0 \times 10^{-3} \text{ M/s}$. At what rate does O_2 form? Give the number in units of 10^{-3} M/s .

Q14 CR2-LO4

Which rate factor explains each observation?

Options: surface area · temperature · concentration

Antacid powder works faster than a whole tablet

Milk spoils faster outside the fridge

Mg fizzes faster in 2 M HCl than in 1 M HCl

Desiccant sachets in tablet bottles

Q15 CR2-LO4

Raising the temperature increases the rate mainly because:

- A. E_a becomes smaller
- B. a much larger fraction of collisions have energy $\geq E_a$
- C. ΔH becomes more negative
- D. more reactant is made

Part 5 · Catalysts and enzymes

Q16 CR2-LO5

A catalyst makes an exothermic reaction release more heat.

True / False

Q17 CR2-LO5

Above about 40 °C, the rate of most human enzyme reactions falls because:

- A. the substrate evaporates
- B. the enzyme denatures and its active site loses its shape
- C. collisions become less frequent
- D. ΔH changes sign

Q18 CR2-LO5

Aspirin and ibuprofen slow down the COX enzymes. Drugs that act this way are called enzyme _____.

Part 6 • Equilibrium and Le Châtelier

For questions 1–4 use $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$, $\Delta H = -92 \text{ kJ}$.

Q19 CR2-LO6

Predict the shift.

Options: right · left · no shift

N_2 is added

NH_3 is removed

The temperature is raised

The pressure is increased

An iron catalyst is added

Q20 CR2-LO6

At dynamic equilibrium:

- A. both reactions have stopped
- B. the forward and reverse rates are equal
- C. reactant and product concentrations are equal
- D. only the forward reaction runs

Q21 CR2-LO6

$\text{Hb} + \text{O}_2 \rightleftharpoons \text{HbO}_2$. In the tissues, where cells use up O_2 , the equilibrium:

- A. shifts right and Hb picks up O_2
- B. shifts left and HbO_2 releases O_2
- C. does not move
- D. stops

Pharmacy scenario · Hydrogen peroxide antiseptic

Scenario

A 3% hydrogen peroxide solution is sold in an **opaque** bottle labelled *Store in a cool place*. It slowly decomposes: $2\text{H}_2\text{O}_2(\text{aq}) \rightarrow 2\text{H}_2\text{O}(\text{l}) + \text{O}_2(\text{g})$, $\Delta H = -196 \text{ kJ}$. ($\text{H} = 1.01$, $\text{O} = 16.00$)

Q22 CR2-LO1

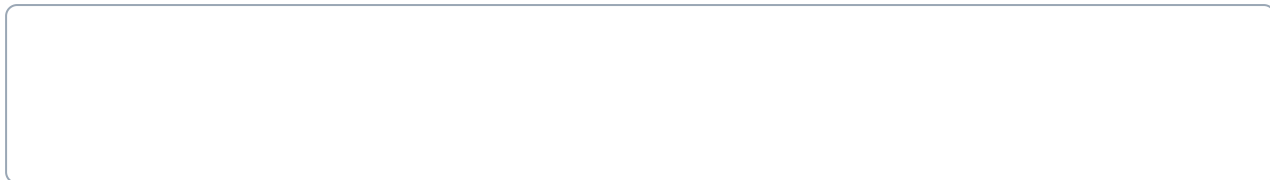
How much heat is released when 3.40 g of H_2O_2 decomposes?

Q23 CR2-LO4

Explain the two storage instructions (opaque bottle, cool place) using collision theory.

Q24 CR2-LO5

On a wound it fizzes immediately. Explain, and sketch on one energy diagram the path with and without the catalyst.



Dr. NAM · Faculty of Pharmacy, UiTM Bertam, Penang

Adapted from *The Basics of GOB Chemistry* (Ball, Hill & Scott), LibreTexts, CC BY-NC-SA 3.0, sections 6.5 and 7.4, with rates and equilibrium expanded from *Introductory Chemistry* (LibreTexts), chapter 15. Shared under the same licence for non-commercial teaching.

Periodic table · each box: atomic number, symbol, atomic mass, **electronegativity** · colours: metal, metalloid, nonmetal, noble gas

1A 1	2A 2	3	4	5	6	7	8	9	10	11	12	3A 13	4A 14	5A 15	6A 16	7A 17	8A 18
1 H 1.01 2.1																	2 He 4.00 —
3 Li 6.94 1.0	4 Be 9.01 1.5											5 B 10.81 2.0	6 C 12.01 2.5	7 N 14.01 3.0	8 O 16.00 3.5	9 F 19.00 4.0	10 Ne 20.18 —
11 Na 22.99 0.9	12 Mg 24.31 1.2											13 Al 26.98 1.5	14 Si 28.09 1.8	15 P 30.97 2.1	16 S 32.07 2.5	17 Cl 35.45 3.0	18 Ar 39.95 —
19 K 39.10 0.8	20 Ca 40.08 1.0	21 Sc 44.96 1.3	22 Ti 47.87 1.5	23 V 50.94 1.6	24 Cr 52.00 1.6	25 Mn 54.94 1.5	26 Fe 55.84 1.8	27 Co 58.93 1.9	28 Ni 58.69 1.9	29 Cu 63.55 1.9	30 Zn 65.38 1.6	31 Ga 69.72 1.6	32 Ge 72.63 1.8	33 As 74.92 2.0	34 Se 78.97 2.4	35 Br 79.90 2.8	36 Kr 83.80 —
37 Rb 85.47 0.8	38 Sr 87.62 1.0	39 Y 88.91 1.2	40 Zr 91.22 1.4	41 Nb 92.91 1.6	42 Mo 95.95 1.8	43 Tc [98] 1.9	44 Ru 101.07 2.2	45 Rh 102.91 2.2	46 Pd 106.42 2.2	47 Ag 107.87 1.9	48 Cd 112.41 1.7	49 In 114.82 1.7	50 Sn 118.71 1.8	51 Sb 121.76 1.9	52 Te 127.60 2.1	53 I 126.90 2.5	54 Xe 131.29 —
55 Cs 132.91 0.7	56 Ba 137.33 0.9	57–71	72 Hf 178.49 1.3	73 Ta 180.95 1.5	74 W 183.84 1.7	75 Re 186.21 1.9	76 Os 190.23 2.2	77 Ir 192.22 2.2	78 Pt 195.08 2.2	79 Au 196.97 2.4	80 Hg 200.59 1.9	81 Tl 204.38 1.8	82 Pb 207.2 1.9	83 Bi 208.98 1.9	84 Po [209] 2.0	85 At [210] 2.2	86 Rn [222] —
87 Fr [223] 0.7	88 Ra [226] 0.9	89–103	104 Rf [267] —	105 Db [268] —	106 Sg [269] —	107 Bh [270] —	108 Hs [277] —	109 Mt [278] —	110 Ds [281] —	111 Rg [282] —	112 Cn [285] —	113 Nh [286] —	114 Fl [289] —	115 Mc [290] —	116 Lv [293] —	117 Ts [294] —	118 Og [294] —
Lanthanides		57 La 138.91 —	58 Ce 140.12 —	59 Pr 140.91 —	60 Nd 144.24 —	61 Pm [145] —	62 Sm 150.36 —	63 Eu 151.96 —	64 Gd 157.25 —	65 Tb 158.93 —	66 Dy 162.50 —	67 Ho 164.93 —	68 Er 167.26 —	69 Tm 168.93 —	70 Yb 173.05 —	71 Lu 174.97 —	
Actinides		89 Ac [227] 1.1	90 Th 232.04 1.3	91 Pa 231.04 1.4	92 U 238.03 1.4	93 Np [237] —	94 Pu [244] —	95 Am [243] —	96 Cm [247] —	97 Bk [247] —	98 Cf [251] —	99 Es [252] —	100 Fm [257] —	101 Md [258] —	102 No [259] —	103 Lr [262] —	

Atomic masses to 2 d.p. ([n] = mass number of the most stable isotope). Electronegativity: Pauling values as in Ball, *The Basics of GOB Chemistry*, Figure 4.4.3 (La–Lu 1.0–1.2; noble gases have no value).

After class: check your understanding

Dr. NAM · Faculty of Pharmacy, UiTM Bertam, Penang

About 25 minutes. Questions follow the six parts of the lecture and end with a pharmacy scenario.

Answer key

Part 1 · Energy changes in reactions

Q1 CR2-LO1

In an exothermic reaction:

- A. the bonds formed release more energy than the bonds broken absorb**
- B. the bonds broken absorb more energy than the bonds formed release
- C. no bonds are broken
- D. the surroundings get colder

More energy is released on forming bonds than absorbed on breaking them, so heat is left over and $\Delta H < 0$.

Q2 CR2-LO1

$2\text{H}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{H}_2\text{O}(\text{l})$, $\Delta H = -572 \text{ kJ}$. What is ΔH for $\text{H}_2(\text{g}) + \frac{1}{2}\text{O}_2(\text{g}) \rightarrow \text{H}_2\text{O}(\text{l})$? Include the sign.

-286 kJ

Halving the coefficients halves ΔH : $-572 \div 2 = -286 \text{ kJ}$.

Q3 CR2-LO1

$\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$, $\Delta H = -890 \text{ kJ}$. How much heat is released when 8.00 g of CH_4 burns? (C = 12.01, H = 1.01)

444 kJ

$8.00 \div 16.05 = 0.498 \text{ mol}$; $0.498 \times 890 = 444 \text{ kJ}$ released.

Q4 CR2-LO1

A cold pack contains 20.0 g of NH_4NO_3 (80.06 g/mol). Dissolving it has $\Delta H = +25.7$ kJ/mol. How much heat does it absorb?

6.42 kJ

$20.0 \div 80.06 = 0.250$ mol; $0.250 \times 25.7 = 6.42$ kJ absorbed (endothermic).

Q5 CR2-LO1

$\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 \rightarrow 6\text{CO}_2 + 6\text{H}_2\text{O}$, $\Delta H = -2803$ kJ. How much heat does the body release by oxidising 4.50 g of glucose (180.18 g/mol)?

70 kJ

$4.50 \div 180.18 = 0.0250$ mol; $0.0250 \times 2803 = 70.0$ kJ released.

Part 2 • Energy diagrams

Q6 CR2-LO2

E_a for the forward reaction:

100 kJ

$160 - 60 = 100$ kJ.

Q7 CR2-LO2

ΔH for the reaction (include the sign):

-35 kJ

$25 - 60 = -35$ kJ; exothermic.

Q8 CR2-LO2

E_a for the reverse reaction:

135 kJ

$160 - 25 = 135$ kJ.

Q9 CR2-LO2

Match each feature of an energy diagram.

the top of the hill → transition state; peak – reactants → E_a ; products – reactants → ΔH ; products below reactants → exothermic

Part 3 • Collision theory

Q10 CR2-LO3

Will each collision form products?

Enough energy, correct orientation → effective; Enough energy, wrong orientation → ineffective; Too little energy, correct orientation → ineffective

Both conditions are needed: energy $\geq E_a$ and the right orientation.

Q11 CR2-LO3

A reaction with a large E_a is slow because:

- A. its particles never collide
- B. only a small fraction of collisions have energy $\geq E_a$**
- C. it is always endothermic
- D. its ΔH is large

Higher E_a means a smaller high-energy tail, so fewer effective collisions.

Part 4 • Reaction rates

Q12 CR2-LO4

In a reaction, [A] falls from 0.50 M to 0.30 M in 40 s. What is the average rate of loss of A? Give the number in units of 10^{-3} M/s.

5×10^{-3} M/s

$0.20 \text{ M} \div 40 \text{ s} = 5.0 \times 10^{-3} \text{ M/s}$.

Q13 CR2-LO4

For $2\text{H}_2\text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{O}_2$, H_2O_2 is used up at $6.0 \times 10^{-3} \text{ M/s}$. At what rate does O_2 form? Give the number in units of 10^{-3} M/s .

$3 \times 10^{-3} \text{ M/s}$

Mole ratio $\text{O}_2 : \text{H}_2\text{O}_2 = 1 : 2$, so $3.0 \times 10^{-3} \text{ M/s}$.

Q14 CR2-LO4

Which rate factor explains each observation?

Antacid powder works faster than a whole tablet → surface area; Milk spoils faster outside the fridge → temperature; Mg fizzes faster in 2 M HCl than in 1 M HCl → concentration; Desiccant sachets in tablet bottles → concentration

Desiccants lower the concentration of water available to hydrolyse the drug.

Q15 CR2-LO4

Raising the temperature increases the rate mainly because:

A. E_a becomes smaller

B. a much larger fraction of collisions have energy $\geq E_a$

C. ΔH becomes more negative

D. more reactant is made

Temperature does not change E_a or ΔH .

Part 5 • Catalysts and enzymes

Q16 CR2-LO5

A catalyst makes an exothermic reaction release more heat.

True / False → **False**

A catalyst lowers E_a only; ΔH is unchanged.

Q17 CR2-LO5

Above about 40 °C, the rate of most human enzyme reactions falls because:

- A. the substrate evaporates
- B. the enzyme denatures and its active site loses its shape**
- C. collisions become less frequent
- D. ΔH changes sign

Heat unfolds the protein, so the substrate no longer fits the active site.

Q18 CR2-LO5

Aspirin and ibuprofen slow down the COX enzymes. Drugs that act this way are called enzyme _____.

inhibitors

Enzyme inhibitors reduce the rate of an enzyme-catalysed reaction.

Part 6 · Equilibrium and Le Châtelier

Q19 CR2-LO6

Predict the shift.

N_2 is added $\rightarrow$ right; NH_3 is removed $\rightarrow$ right; The temperature is raised $\rightarrow$ left; The pressure is increased $\rightarrow$ right; An iron catalyst is added $\rightarrow$ no shift

Raising T favours the endothermic (reverse) direction; higher pressure favours fewer gas moles (2 vs 4).

Q20 CR2-LO6

At dynamic equilibrium:

- A. both reactions have stopped
- B. the forward and reverse rates are equal**
- C. reactant and product concentrations are equal
- D. only the forward reaction runs

Both reactions continue at equal rates, so concentrations stay constant.

Q21 CR2-LO6

$\text{Hb} + \text{O}_2 \rightleftharpoons \text{HbO}_2$. In the tissues, where cells use up O_2 , the equilibrium:

- A. shifts right and Hb picks up O_2
- B. shifts left and HbO_2 releases O_2**
- C. does not move
- D. stops

Removing O_2 shifts the equilibrium left to replace it.

Pharmacy scenario · Hydrogen peroxide antiseptic

Q22 CR2-LO1

How much heat is released when 3.40 g of H_2O_2 decomposes?

9.79 kJ

$M = 34.02 \text{ g/mol}$; $3.40 \div 34.02 = 0.0999 \text{ mol}$. ΔH is for 2 mol, so $98 \text{ kJ/mol} \times 0.0999 = 9.79 \text{ kJ}$ released.

Q23 CR2-LO4

Explain the two storage instructions (opaque bottle, cool place) using collision theory.

Opaque bottle: light supplies energy that helps molecules reach E_a , so blocking light slows photodecomposition. **Cool place:** at lower temperature, molecules collide less often and a smaller fraction of collisions have energy $\geq E_a$, so decomposition is slower and the product lasts until its expiry date.

Q24 CR2-LO5

On a wound it fizzes immediately. Explain, and sketch on one energy diagram the path with and without the catalyst.

Catalase in blood and tissue is an enzyme (catalyst). It provides a pathway with a much lower E_a , so many more collisions are effective and O_2 bubbles form at once. Diagram: same reactant and product levels (same $\Delta H = -196 \text{ kJ}$), but a lower peak for the catalysed path.

Dr. NAM · Faculty of Pharmacy, UiTM Bertam, Penang

Adapted from *The Basics of GOB Chemistry* (Ball, Hill & Scott), LibreTexts, CC BY-NC-SA 3.0, sections 6.S and 7.4, with rates and equilibrium expanded from *Introductory Chemistry* (LibreTexts), chapter 15. Shared under the same licence for non-commercial teaching.